Exercise 10: Mobile and Accessible Dashboards

One dashboard, two layouts, three captions. No memo.

Item	Details
Points	100
Data	clean/mvu_student_dim.csv
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 8 (dashboards and device layouts) and Ch 11 (dynamic dashboards). Tableau Help: Create Dashboard Layouts for Different Device Types; Show or Hide Dashboard Objects; Create Views in Tooltips. WebAIM contrast checker (webaim.org/resources/contrastchecker).
Version	Dynamic zone visibility needs Tableau Desktop 2022.3 or later. On an older version, use a Show/Hide button instead (step 5 of Worksheet 2).
Starter workbook	MIS6380_Exercise10_Starter.twbx: data connected, student_id set as a Dimension, and the four measures ready. You add the 2020–21 cohort filter in Set up.
Walkthrough	Module 10 of the MVU Tableau Walkthrough follows these same steps, with checkpoints.
Submit	One packaged workbook (.twbx) with the dashboard 10 Dashboard (Default and Phone layouts) and a Captions dashboard. No memo.

Before you start

In Exercise 5 you built a dashboard for one screen and one kind of reader: someone at a desk with a mouse. Many readers will open your work on a phone, where nothing hovers and a 1,200-pixel layout becomes a smudge. Others can't tell red from green, or need larger text. This exercise rebuilds one dashboard so it works for all of them.

Set up

1. Open the Exercise 10 starter workbook, or connect mvu_student_dim.csv and set student_id to a Dimension.
2. Check (or create) **Students** = COUNTD([student_id]); **Graduation Rate** = AVG([is_graduated]); **Withdrawal Rate** = AVG([is_withdrawn]) (both Percentage, 1 decimal); **Average Aid** = AVG([total_aid_awarded]) (Currency, 0 decimals).
3. Filter cohort_year to 2020 and 2021 and apply it to all worksheets using this data source (right-click the filter > Apply to Worksheets > All Using This Data Source).

Checkpoint: How many students are in the 2020–21 cohorts, and what is their graduation rate?

Worksheet 1: a desktop layout, then a phone layout

1. Sheet **KPIs**: Measure Names on Columns, Measure Values on Text, showing only Students, Graduation Rate, Withdrawal Rate and Average Aid. Make the numbers large (about 24 pt).

2. Sheet **By College**: college on Rows, Graduation Rate on Columns, sorted descending, labels on. Sheet **By Intensity**: enrollment_intensity on Rows, Withdrawal Rate on Columns, labels on.

Checkpoint: What are the withdrawal rates for part-time and full-time students?

3. Dashboard **10 Dashboard**, Size Fixed 1200 × 800. Use containers: a title, the KPIs across the top, By College on the left (about two-thirds of the width) and By Intensity on the right.
4. Click **Device Preview**, choose **Phone**, then **Add Phone Layout**. Set the layout to Fit Width.
5. Edit the phone layout: KPIs first, then By College, then By Intensity, each at a height a thumb can scroll past. Remove anything a phone reader doesn't need, such as a legend that repeats a label or a long text block.
6. Preview at two phone models in Device Preview. Check that every label is readable without zooming.

Checkpoint: What do the Withdrawal Rate and Average Aid cards read?

Worksheet 2: details on demand, with or without a mouse

1. Sheet **Programs**: program_name on Rows, Graduation Rate on Columns, sorted descending, labels on.
2. **Viz in Tooltip:** on By College, open Tooltip > Insert > Sheets > Programs. Hover over each college on the desktop layout.

Checkpoint: Hovering over College of Health Sciences, which program has the highest graduation rate, and what is it?

3. **Dynamic zone visibility:** create a Boolean parameter **Show Programs** (current value False) and a calculated field **Show Programs Zone** = [Show Programs]. Add Programs to the dashboard in its own container below the charts. Select the container, open the Layout pane and tick **Control visibility using value**: Show Programs Zone. Show the parameter control and label it "Show program detail".
4. No dynamic zone visibility in your version? Make the Programs container floating, then select it and choose **Add Show/Hide Button**.
5. Add the Programs container and its control to the phone layout too, since phone readers can't hover.

Checkpoint: Which program has the lowest graduation rate in the 2020–21 cohorts?

Worksheet 3: color, contrast and type

1. Check your colors in the WebAIM contrast checker: UTD orange (#E87500) and UTD green (#154734), each on white.

Checkpoint: What is the contrast ratio of UTD orange on white, to one decimal? And UTD green on white?

2. WCAG AA asks for 4.5:1 for normal text and 3:1 for large text and chart marks. Use orange only where it passes for that use, for example a highlighted bar, never body text or small labels.
3. If any chart uses color for categories, switch to Tableau's **Color Blind** palette and label the marks directly, so color is never the only cue.
4. Make every label at least 10 pt on the phone layout, and rewrite each sheet title as a finding with its cohorts, for example "Part-time students withdraw at 3.6 times the full-time rate (2020–21)".

5. Test it: take a screenshot of the dashboard and run it through a color-blindness simulator such as Coblis, or view the published dashboard with Chrome DevTools > Rendering > Emulate vision deficiencies.

Your three captions

Captions don't belong on a phone screen, so put them on a separate dashboard named **Captions**, in three Text objects: one or two sentences answering each question.

Worksheet	What your caption must answer
Worksheet 1	What did you change, remove or reorder for the phone layout, and why? What would a phone reader miss if you had only shrunk the desktop layout?
Worksheet 2	How does a reader get program detail on a desktop, and how on a phone? Why isn't Viz in Tooltip enough on its own, and what do the program bars show that the college bars hide?
Worksheet 3	Where did you use UTD orange, and where did the contrast check stop you? How does a reader who can't tell colors apart read your dashboard?

Worksheet 2's caption carries 25% on its own. Describing the tooltip without saying it fails on a touch screen does not earn full marks. Name both paths to the detail, and name a program the college view hides.

Before you submit: open Device Preview once more and check the phone layout from top to bottom.

Submit for Exercise 10

- Sheet names: KPIs · By College · By Intensity · Programs, plus the dashboards 10 Dashboard and Captions
- Worksheet 1: a fixed-size desktop layout built with containers, and an edited phone layout
- Worksheet 2: a Viz in Tooltip on By College, and a Programs container shown by a parameter (or a Show/Hide button) on both layouts
- Worksheet 3: contrast-checked colors, direct labels, finding titles and readable text on the phone
- A Captions dashboard with three captions
- Packaged workbook (.twbx). No memo.

Optional extensions

- Publish to Tableau Public and open the dashboard on your own phone. What looked different from Device Preview?
- Add navigation buttons between 10 Dashboard and Captions, and set a sensible tab order for keyboard users.